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Chapter 11

GELFAND-P000705

5/IV-77 Gelfand Sem. (34th year of activity).



Original source details: GELFAND-P000705

Uncertain reading. The second Roman mark has two converging strokes. The month reads IV.

Possible seminar topic: noncommutative algebraic geometry (drawing on recent work by I. N. Bernstein, I. M. Gelfand, S. I. Gelfand

V. Drinfeld. On Atiyah's work on the Yang–Mills eqs.

A 4-dimensional sphere is given S^4, with a smooth complex vector bundle ξ. A connection ∇ on this bundle.

The pair satisfies the eqs. of duality, if

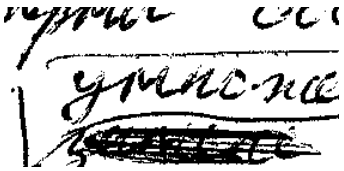
$$*K = K \tag{1}$$

(K — the curvature form of this connection)

K — a form with values in endomorphisms of the bundle

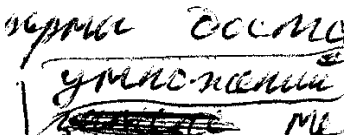
$*$ — is constructed from the standard metric on S^4; either orientation may be fixed.

To construct $*$ from 2-forms to 2-forms, it is enough to have a conformal structure (on [crossed out: replacing?] [inserted above: multiplying] the metric



Original source details: GELFAND-P000705

Undeciphered passage. The crossed-out word begins with a stroke like "з." One possible reading is "замене" (replacing).



Original source details: GELFAND-P000705

Uncertain reading. "Multiplying" is circled above the line. A stroke from it points to the crossed-out word.

[source number: 40]

by a number leaves it unchanged)

The fibres also have a metric (Hermitian). The connection preserves the metric on the fibre.

The connection is a classical field, whose quantum analogue describes the gluons that are supposed to hold the quarks together.

What does one do next with a solution? Quantization and so on. To the pair (ξ, ∇) corresponds a linear operator, from the spectrum, extract a number (renormalized determinant). Then integrate over the set of solutions. This is the denominator.



Uncertain reading. The abbreviation before "number" tentatively reads "извл." (extract).

Original source details: GELFAND-P000706

There is a one-to-one corresp. between solutions and certain alg. bundles on $\mathbb{C}P^3$. Consider the projection $\mathbb{C}P^3 \to S^4$.

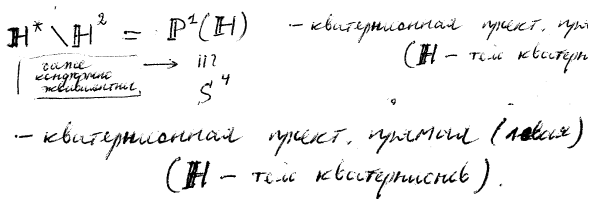


Diagram and typed description.

$\mathbb{H}^*\backslash\mathbb{H}^2 =$ $\mathbb{P}^1(\mathbb{H})$

they are conformally equivalent $\longrightarrow$ $\wr\|$ S^4

— the quaternionic projective line (left) ($\mathbb{H}$ — the division ring of quaternions).

Original source details: GELFAND-P000706

$\mathbb{C} \subset \mathbb{H}$, hence a surj. $\mathbb{C}^*\backslash\mathbb{H}^2 = \mathbb{C}P^3$ and we have a map $\mathbb{C}P^3 \to S^4$ (its fibres are two-dimensional).

Conf. structure on $\mathbb{P}^1(\mathbb{H})$ — with a point removed, it is $\mathbb{R}^4 \cong \mathbb{H}$ with the standard conf. structure.



Uncertain reading. The superscript before the parenthesis is read as 1, as in the formula above the diagram. The lower stroke approaches the parenthesis.

Original source details: GELFAND-P000706

Conf. structure: at $\forall$ points a metric is given up to a positive multiplier. The fibres of $\mathbb{C}P^3 \to S^4$ are $\mathbb{C}P^1$.

Atiyah's thm. establishes an isomorphism between

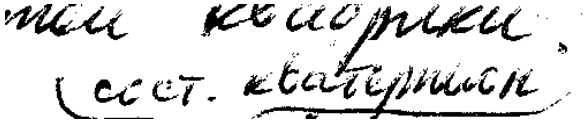
1) pairs ξ, ∇, satisfying the eqs. of duality

2) alg. vect. bundles on $\mathbb{C}P^3$, trivial on each fibre of the map $\mathbb{C}P^3 \xrightarrow{f} S^4$.

Establishing the isomorphism:

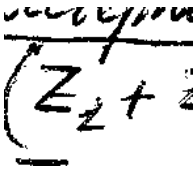
All lines (proj. complex) in $\mathbb{C}P^3$ form a quadric, the fibres of the map f — the real points of this quadric. In $\mathbb{C}P^3$ there is an involution (z_1, z_2, z_3, z_4);

form a quaternion
$$(z_1 + z_2 j,\ z_3 + z_4 j)$$



Original source details: GELFAND-P000706

Uncertain reading. The circled note above the pair tentatively reads "сост. кватернион" (form a quaternion).



Original source details: GELFAND-P000706

Uncertain reading. The first index in the pair is read as 1. The slanting stroke and broad foot also allow 2.

Multiply on the left by j:
$$\bar{z}_1 j - \bar{z}_2, \qquad \bar{z}_3 j - \bar{z}_4$$

GELFAND-P000707

Involution:
$$(z_1, z_2, z_3, z_4) \mapsto (-\bar{z}_2, \bar{z}_1, -\bar{z}_4, \bar{z}_3)$$

Take the complex lines invariant under this involution. This gives S^4.

Triviality is understood in the alg. ($\sim$ complex) sense.

The bundles are pulled back from S^4, but on S^4 there is no complex structure.

1) $\Rightarrow$ 2)
$$\xi \mapsto f^* \xi$$

Details next time.

I. Krichever. On constructing exact solutions of other nonlinear equations.

$$\left[\frac{\partial}{\partial y} - L_1, \frac{\partial}{\partial t} - L_2\right] = 0 \tag{1}$$

$$L_1 = \sum_{i=0}^{n} u_i(x, y, t) \frac{d^i}{dx^i} \tag{2}$$



Original source details: GELFAND-P000707

Undeciphered passage. Two vertical strokes with transverse traces are visible in the right margin beside (2).

$$L_2 = \sum_{j=0}^{m} v_j(x,y,t)\frac{d^j}{dx^j} \tag{3}$$

The coefficients may also be matrices.

There are solutions in general position.

Separatrix family — multisoliton [crossed out] solutions, of the eq.

$$u_t = 6uu_x + u_{xxx}.$$



Undeciphered passage. A short passage between "multisoliton" and "solutions" is heavily crossed out.

Original source details: GELFAND-P000707

There are parameters. They are almost always conditionally periodic.

For certain vals. of the parameter — rational.

Manin. Solutions of alg. type (without y) — those that are common solutions of (1) and

$$[L_1, L_3] = 0 \tag{4}$$



Undeciphered passage. An isolated T-shaped mark is visible to the right of (4).

Original source details: GELFAND-P000707

(L_3 — aux. operator)

Solns. [crossed out] of alg. type — such that $\exists L_3 \notin \mathbb{C}[L_1, L_2]$, such that cond. is satisfied (4).

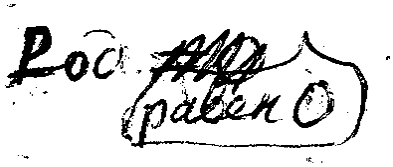


Undeciphered passage. A short passage between "Solns." and "of alg. type" is crossed out.

Original source details: GELFAND-P000707

Degenerate solutions: in the lin. space of lin. combinations of coeffs. of fractional powers, to each corresponds an alg. curve.

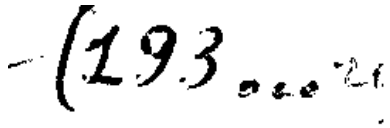
Its genus is constant almost everywhere. Degeneration — where the genus changes. The genus [crossed out $\neq 0$?] [boxed below the line: equals 0] where there are multisoliton solutions and something else.



Uncertain reading. The crossed-out writing resembles $\neq 0$. "Equals 0" is enclosed below.

Original source details: GELFAND-P000707

There is alg. dependence of L_1 and L_2 — Jacobson's thm. (193...



Uncertain reading. Three low marks resembling an ellipsis follow "193," with faint strokes at the right edge.

Original source details: GELFAND-P000707

GELFAND-P000708

[source number: 41]

Moser's work: [above the line: (Mc-Kean and Airault)] seek solns. rational in x of KdV. In t one obt. ordinary eqs. that can be solved explicitly.

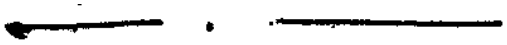


Diagram and typed description.

$\longleftarrow \bullet \longrightarrow$

Original source details: GELFAND-P000708

The rationality cond. for KdV replaces the 3rd operator.

Apparently the same holds for $\forall$ of the Lax eqs.:

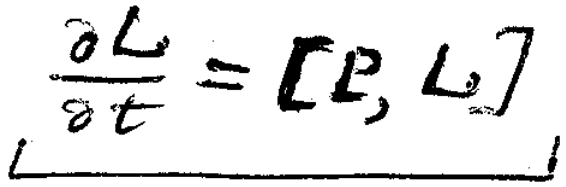


Diagram and typed description.

$$\underbrace{\frac{\partial L}{\partial t} = [P, L]}$$

Original source details: GELFAND-P000708

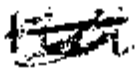
GELFAND-P000709

12/IX-77 Gelfand Sem.

G. Margulis.

G — a semisimple connected Lie group, for simplicity let its centre be trivial, $G \supset \Gamma$ — a discrete subgroup, such that [crossed out]

$$\nu(G/\Gamma) < +\infty \qquad (\nu \text{ — volume induced by Haar measure})$$

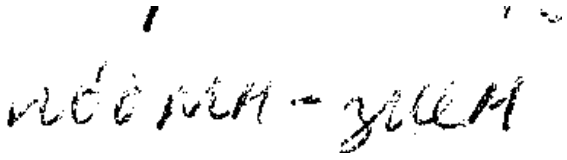


Undeciphered passage. A short passage to the left of the formula is crossed out.

Original source details: GELFAND-P000709

Such Γ are called lattices.

An algebraic group G — a subgroup of $GL(n, \mathbb{C})$, that is an alg. variety in $GL(n, \mathbb{C})$.



Uncertain reading. The abbreviation after "alg." tentatively reads "много-зием" (variety).

Original source details: GELFAND-P000709

k-group — a group def. over the field k (given by eqs. with coeffs. in k). (here $k \subset \mathbb{C}$ a subfield) (Otherwise: it is called k-closed).

Let G — $\mathbb{Q}$-group.

$G(k)$ — set of matrices with coeffs. in k, $\in$ our set.

One can take points over a subring of $\mathbb{C}$

$$G(\mathbb{R}) \supset G(\mathbb{Z})$$

$G(\mathbb{Z})$ is a group if [crossed out: $G(\mathbb{R})$] $G(\mathbb{Z}) \subset SL(n, \mathbb{Z})$.



Original source details: GELFAND-P000709

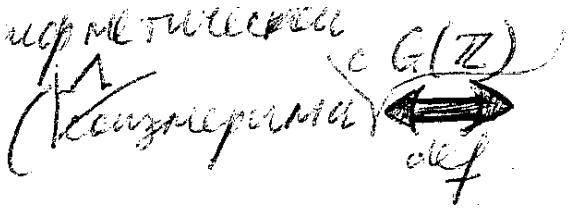
Uncertain reading. The crossed-out writing before $G(\mathbb{Z}) \subset SL(n, \mathbb{Z})$ resembles $G(\mathbb{R})$.

1) $G(\mathbb{Z})$ is called an arithm. subgroup.

2) $\Lambda \subset G(\mathbb{R})$

if Λ is commensurable with $G(\mathbb{Z})$, then Λ is also called arithmetic.

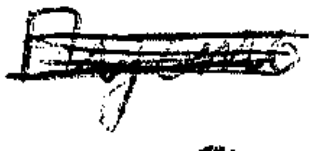
([above the line: Λ] commensurable [above the line: with $G(\mathbb{Z})$] $\underset{\text{def}}{\Longleftrightarrow}$ $\Lambda \cap G(\mathbb{Z})$ has finite index in Λ and in $G(\mathbb{Z})$)



Original source details: GELFAND-P000709

Uncertain reading. Λ and "with $G(\mathbb{Z})$" are inserted above the line. "def" stands under the double arrow.

[crossed out: Question]

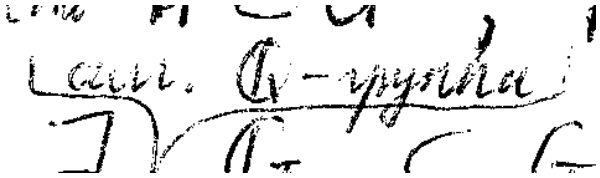


Original source details: GELFAND-P000709

Uncertain reading. Several rules cross out a word that tentatively reads "Question."

Let $H \subset G$, H is called an arithmetic subgroup, if [above the line with a pointer: alg. $\mathbb{Q}$-group]

$$\exists G \subset GL(n, \mathbb{C}) \quad \text{and} \quad f : G(\mathbb{R}) \longrightarrow G,$$



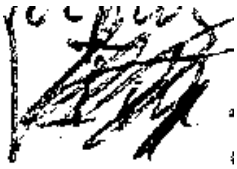
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Uncertain reading. "alg. $\mathbb{Q}$-group" is written above with a pointer toward the following G.

such that

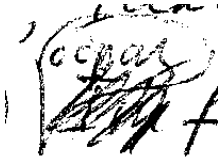
1) f is an epimorphism, 2) Ker f is compact.

3) [above the line with a pointer: (image)] [crossed out] $f(G(\mathbb{Z}))$ is commensurable with H.



Original source details: GELFAND-P000709

Undeciphered passage. Writing before $f(G(\mathbb{Z}))$ is crossed out with slanting strokes.



Original source details: GELFAND-P000709

Uncertain reading. "Image" is inserted above the crossed-out writing.

GELFAND-P000710

Property: if H — an arithm. subgroup, then $\nu(G/H) < \infty$.

(Borel–Harish-Chandra theorem

Example: $SL(n,\mathbb{R}) \supset SL(n,\mathbb{Z})$

$$\nu\big(SL(n,\mathbb{R})/SL(n,\mathbb{Z})\big) < +\infty$$

(Minkowski) [crossed out: or Hermite?].



Original source details: GELFAND-P000710

Uncertain reading. Two words after "Minkowski" are crossed out. They tentatively read "or Hermite."

Exercise: do it for $n = 2$ — this is equiv. to finding the area of a triangle in the Lobachevsky plane.



Original source details: GELFAND-P000710

Diagram and typed description.



Theorem: under certain additional conds. the converse holds, i.e., every lattice is arithmetic. Proof is in the appendix to the book by Raghunathan.

For $SL(2,\mathbb{R})$ it is false — there are parameters for tiling the Lobachevsky plane, and only countably many arithm. subgroups.

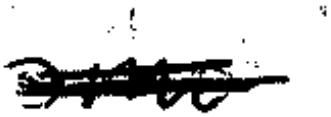
Selberg's conjecture: it is true for the remaining groups Siegel also worked on this.

$$G = \prod G_i, \qquad G_i \text{ — simple Lie groups}$$

A latt. Γ is called reducible, if $G = G_1 \times G_2$, $\dim G_i > 0$, and $(G_1 \cap \Gamma) \times (G_2 \cap \Gamma)$ has finite index in Γ.

We study irreducible lattices.

Conjecture: for an irred. subgroup in $G \neq SL(2,\mathbb{R})/\{\pm E\}$ [crossed out: this?] $\forall$ lattice is arithmetic.



Uncertain reading. The crossed-out word before $\forall$ tentatively reads "это" (this).

Original source details: GELFAND-P000710

Makarov, Vinberg: found nonarithm. subgrps. in groups of motions of Lobachevsky spaces $SO(3,1)$, $SO(4,1)$, $SO(5,1)$).

Answer: if

$$\operatorname{rank} G \geq 2, \quad \text{then the theorem holds.}$$

There is a parameter only in $SL(2,\mathbb{R})$. In the remaining cases rigidity. (Selberg, A. Weil)

In the other groups of rank 1 it is unknown.

Corollary of the rigidity thm. (Selberg): $\forall$ lattice [crossed out] in $SL(n,$ [crossed out] $\mathbb{R})$ is conjugate to a lattice with algebraic coefficients.



Undeciphered passage. A short passage after "lattice" is heavily crossed out.

Original source details: GELFAND-P000710



Undeciphered passage. An isolated sign is crossed out before $\mathbb{R}$ in $SL(n,\mathbb{R})$.

Original source details: GELFAND-P000710

GELFAND-P000711

[source number: 44]

Main lemma.

Let G — a semisimple Lie group, [enclosed insertion above the line: rank $G \geq 2$], Γ — [enclosed insertion above the line: an irreducible] lattice in G, [inserted between the lines: G has no compact factors], k — a local field of char. 0, i.e. $\mathbb{C}$, $\mathbb{R}$ or a finite extension of $\mathbb{Q}_p$, $n \in \mathbb{N}$ and let there be given a homomorphism $T : \Gamma \longrightarrow GL(n, k)$. Suppose

1) $\overline{T(\Gamma)}$ is connected and simple. (Zariski closure)

2) $T(\Gamma)$ is not rel. compact

Then

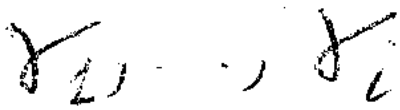
I. $k = \mathbb{C}$ or $\mathbb{R} \Rightarrow T$ extends to a repr. of the group G

II. $k = \mathbb{Q}_p \Rightarrow$ conditions 1) and 2) are incompatible

Deduction of the theorem from the lemma.

$G = G(\mathbb{R})$, K — a field of alg. numbers, we may assume $\Gamma \subset G(K)$ (easily proved [inserted above the line: by Selberg's thm.]).

Let $\gamma_1, \ldots, \gamma_i$ be generators.



Uncertain reading. The last subscript is tentatively i, possibly l.

Original source details: GELFAND-P000711

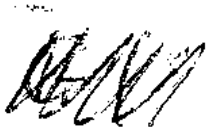
Consid. the denominators of the corresp. matrices and the prime factors occurring in their factorization $p_1, \ldots, p_m$.



Uncertain reading. The last subscript in the list of prime factors is overwritten. It tentatively reads m.

Original source details: GELFAND-P000711

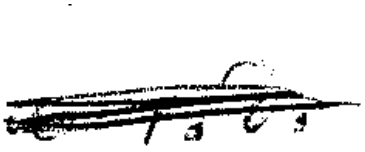
It suff. to prove that the image of



Undeciphered passage. A short passage before Γ is crossed out.

Original source details: GELFAND-P000711

Γ in $G(\mathbb{Q}_p)$ is relat. compact. Then the denominators are bounded



Undeciphered passage. A word after "bounded" is crossed out.

Original source details: GELFAND-P000711

jointly (from the structure of the topology in $\mathbb{Q}_p$).

Compactness follows from assertion II of the lemma. ■

GELFAND-P000712

19/IX—77 Gelfand Sem..

V. Drinfeld. Consid. S^4 with the conformal structure induced by the standard metric.

ξ — a smooth complex vector bundle over S^4.



Uncertain reading. The crossed-out text before the bracket tentatively reads “Кир” (Kir).

Original source details: GELFAND-P000712

[A physicist is present: Kirzhnits, David Abramovich].

Duality equations. A_μ — a matrix-valued fn. of 4 vars.,

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu], \qquad \mu = 1, 2, 3, 4. \tag{1}$$



Uncertain reading. The crossed-out expression before equation (2) tentatively reads F_{ijk}.

Original source details: GELFAND-P000712

$$F_{ij} = \frac{1}{2}\varepsilon_{ijk\ell}F_{k\ell} \qquad \text{(eqns. of duality)} \tag{2}$$

$$[\partial_\mu + A_\mu, F_{\mu\nu}] = 0 \qquad \text{(eqns. of Yang–Mills)} \tag{3}$$

GELFAND-P000713

Kirzhnits David Abramovich (from FIAN).

On certain characteristic ideas of theor. physics.



Uncertain reading. The isolated mark before “Frontier” tentatively reads “4.”

Original source details: GELFAND-P000713

Frontier: 1) elem. particles 2) cosmology 3) transition from dyn. systems to statist.

All this is now closely connected!



Undeciphered passage. A heavily overwritten or crossed-out mark precedes the heading.

Original source details: GELFAND-P000713

Theory of elementary particles.

Arose ≈ 50 years ago.

For the first 20 years QFT ideas predominated, i.e. as in quant. mechanics.

Late 1950s: doubts.

Landau: "The Hamilton. method is a corpse. Our task is to bury it with all the honors due to it!"

1960s — group approach.

Now a renaissance of the field approach. The idea responsible:

1. Spontaneously broken symmetry.

Sensitivity to macroscopic effects (symmetry breaking).

2. Macroscopic effects.

3. History of the universe — an example of $\overbrace{\text{breaking}}^{\text{of such an influence}}$ of spont. symm.

Cosmology

The universe is hot. Why?

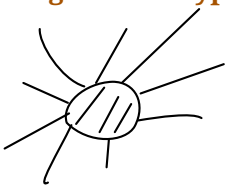
4. "Dynamics — statistics".

First, on pt. 4. A few years ago something fundamentally new was said: Hawking, Bekenstein

a region in phase space spreads out (owing to instability)

Diagram and typed description.





Original source details: GELFAND-P000713

— was known. (e.g. Sinai's billiard)

a syst. with positive entropy.

The language of information (Szilard, 1927, then mathematicians) worked here.

It spread out — we lost information

$$\Delta I = \text{Information} = -\ln \frac{P_f \ (= \text{no. of poss. answers to the question at the end})}{P_i \ (= \text{—}\,''\,\text{— at the beginning})}$$

Entropy

The macrostate is determ. by a small number of parameters. With these, different particle distributions are possible (degeneracy).

Entropy $S = \ln(N_{\text{microscopic}})$

I. M. G.: Entropy: 27 coins, 1 counterf. lighter than the others — 3 weighings.

N coins, 1 counterf. differs in weight. For what N can we find it in 3 weighings:

Solution: we know half as much (13).

N weighings $(3^N - 1)/2$.

GELFAND-P000714

[source number: 47]

Algorithm.

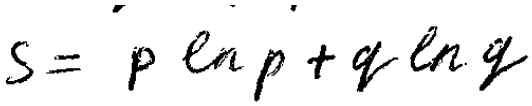
The coins must be divided so that the piles have equal entropy!!

(Divide the entropy into 3 equal parts).

No need for exhaustive search!

coin p — prob. of heads
q — prob. of tails

$$S = p \ln p + q \ln q$$



Uncertain reading. There is no visible minus sign before $p \ln p$ in the manuscript.

Original source details: GELFAND-P000714

Additivity makes it reasonable.

N outcomes $\quad -\sum p_i \ln p_i \qquad$ If $p_i = \frac{1}{N}$, we get $S = \ln N$

Inform. = change in entropy.

Soln. of problem: look at the change in probabilities.

Start: $4 + 4, 5$

$$\Delta I = -\ln \frac{N_f}{N_i}$$

K. D. A. $\quad \Delta S = -\Delta I.$

Example: gas



Diagram and typed description.



initial state final

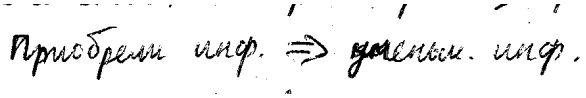
Original source details: GELFAND-P000714

$$S = \ln \frac{V}{N} \qquad \text{(per molecule)}.$$

$$\boxed{\Delta S = \ln 2.}$$

Consequence: loss of inf. $\Rightarrow$ increase in entropy

Gained inf. $\Rightarrow$ decrease in inf.



Uncertain reading. The start of “уменьш.” (decrease) is overwritten. The next abbrev. reads “инф.” (inf.).

Original source details: GELFAND-P000714

Maxwell's demon lowers entropy (keeps track of particles)

Are there any other ways of losing inf.?

Perhaps there are



Undeciphered passage. A word after "there are" is heavily crossed out.

Original source details: GELFAND-P000714

forces retaining information



Undeciphered passage. An isolated mark before "in a g. volume" is crossed out.

Original source details: GELFAND-P000714

in a g. volume.

A known situation where inform. is cut off by phys. means

— inf. does not get out.

There is such a force: gravitation! (and it is the only one)

Long-range and of one sign (!)

"Black hole" — a body that does not let information out.

Laplace thought of it: Speed of light = 2nd cosmic speed.

$$\frac{mv^2}{2} - \gamma\frac{mM}{R} = 0$$

$$\frac{mc^2}{2} = \gamma\frac{mM}{R},$$

$$\boxed{R = \frac{2\gamma M}{c^2} = r_g}$$

At this radius light will not leave the body (nor any signal)

— it turns out to hold in exact science too

GELFAND-P000715

$$R_{\text{Sun}} = 10^6 \text{ km}$$



Uncertain reading. An isolated sign before "For" is heavily crossed out.

Original source details: GELFAND-P000715

For its mass the required rad. is 7 km.

In principle we can know nothing about a black hole except a few params. — — mass, charge,



Original source details: GELFAND-P000715

Undeciphered passage. The entry after "charge" is heavily crossed out.

radius.

As with a gas. An enormous number of microstates. In fact it is a thermodyn. object!!

Hawking: the black hole's surf. S (!) increases in all processes. Merger of 2 black holes.
(earlier)
Geom. clearly increases (from Laplace's formula)

The letter S played a remarkable role.

$$\boxed{\frac{1}{T} = \frac{\partial S}{\partial E}} \quad \Rightarrow \quad T \text{ — the black hole's "temp.".}$$

Bekenstein — removed the quotation marks! Counted the information lost in forming
(Hawking's pupil)
black holes!! Agrees with informational entropy.

Let us throw a hot body into a black hole. It increases the surf. more than a cold one, in exact corresp. with the entropy increase.

[crossed out: Hawking.] Hawking. Merger of pairs of particles. The black hole radiates and its spectrum is Planckian!

$$\frac{1}{e^{\frac{\hbar\omega}{kT}} - 1}$$

T — that temperature.

Inf. is lost not by stochasticization!!, but goes into black holes. (mechanically cut off)

Other ways

Uhnru?

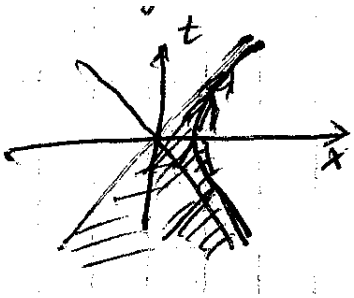


Original source details: GELFAND-P000715

Uncertain reading. The spelling tentatively reads "Uhnru." The passage refers to W. G. Unruh.

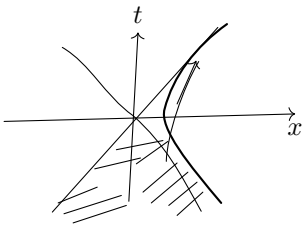
Homog. gravitational field.

Let a particle move with uniform acceleration in vacuum. Light from some places will not reach it.



Original source details: GELFAND-P000715

Diagram and typed description.

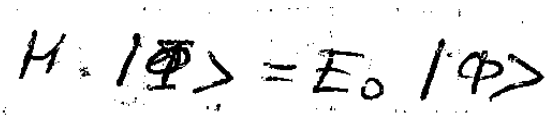


The exact joins of the overlapping strokes and curves at right are indeterminate.

It turns out that light reaches it only from half of space. The obs. will see (calculation) a Planck distribution. The zero-point oscillations look distorted to him. This is a difficult calculation.

H — Ham. of the electromagn. field.

$$H|\Phi\rangle = E_0|\varphi\rangle$$



Original source details: GELFAND-P000715

Uncertain reading. The state sign at left is heavily retraced and looks like Φ. At right it looks like φ.

The state vector from the point of view of a non-inert. obs. $\rightarrow$ Calc. prob. of $1^{\underline{\text{st}}}$, $2^{\underline{\text{nd}}}$ and more

Original source details: GELFAND-P000715

Uncertain reading. Two clipped groups of strokes are visible beyond the right page border.

GELFAND-P000716

[source number: 48?]

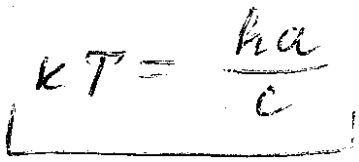


Original source details: GELFAND-P000716

Uncertain reading. The last digit of the folio number tentatively reads 8, possibly 2.

photons.

$$\boxed{kT = \frac{ha}{c}} \quad \Rightarrow \quad \text{temperature}$$



Original source details: GELFAND-P000716

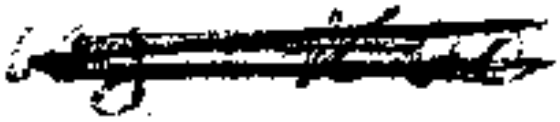
Uncertain reading. The first sign in the numerator reads h. A distinct bar for $\bar{h}$ is indiscernible.

Heating of the gas at the finish.

Why is the universe hot?

[At left] Experiment — it is cooling now ||

[At right] $T = 3°\mathrm{K}$ — from the distrib. of



Undeciphered passage. A word before “Planck” is heavily crossed out.

Original source details: GELFAND-P000716

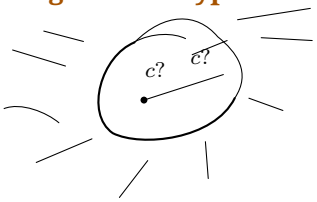
Planck for rad. from space. [below the formula: (relic radiation)]

There was a region of the unknown. It arose at some point in time. Before this it makes no sense [below: to ask].

There is an event horizon.



Diagram and typed description.



The ends of the interior strokes and the two small signs are only partly legible.

Original source details: GELFAND-P000716

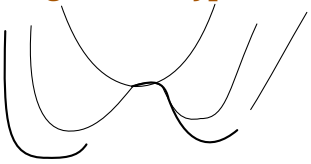
closed region (light has had no time to arrive from there). (⇒ hence this temp. for us.)

—·—

Spontaneous symm. breaking.



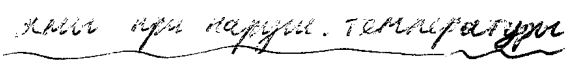
Diagram and typed description.



The precise joins where the lines overlap are indeterminate.

Original source details: GELFAND-P000716

evolution of the well upon a disruption of temperature



Uncertain reading. The caption tentatively reads “evolution of the well upon a disruption of temperature.” “Minima” is also possible for “well.”

Original source details: GELFAND-P000716

—·—

An idea of Senya Vishik, on why the Gelfand sem. is unsterilized: “To integrate successfully over a large domain one must use the Monte Carlo method.”

GELFAND-P000717

[source number: 55]

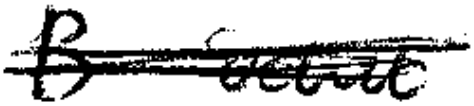
3/X—77 Gelfand Sem..

To be treated systematically: “noncommutative algebraic geometry”

L. A. Dikii. On new Lie algebras.

(Gelf., Dikii, Funct. Anal., 11:2).

Def. 1. The algebra A.



Uncertain reading. The crossed-out text at right begins “В” (In), followed by a word tentatively read as “себе.”

Original source details: GELFAND-P000717

$$u_k^{(i)}, \qquad k = 0, 1, \dots, n; \qquad i = 0, 1, 2, \dots$$

$$A = \left\{ \sum a\, u_{k_1}^{(i_1)} u_{k_2}^{(i_2)} u_{k_3}^{(i_3)} \cdots u_{k_N}^{(i_N)} \right\}, \qquad a \in \underset{\text{(field)}}{K},$$

A — the alg. of polynomials in the generators $u_k^{(i)}$

$$\boxed{A = K[u_k^{(i)}].}$$

$$\xi = \sum a_{k,i} \frac{\partial}{\partial u_k^{(i)}}, \qquad a_{k,i} \in A.$$

(the sum is infinite) ξ is called a derivation

$$\frac{d}{dx} = \sum_{k,i} u_k^{(i+1)} \frac{\partial}{\partial u_k^{(i)}}$$

$$\frac{d}{dx} u_k^{(i)} = u_k^{(i+1)}; \qquad u_k^{(i)} = \left(\frac{d}{dx}\right)^i u_k; \qquad u_k \underset{\text{def}}{=} u_k^{(0)}$$

Def. 2. The space $\widetilde{A}$ (space of functionals)

From now on, always $u_n = 1$, $u_{n-1} = 0$.

$$\left\{ f : \exists g,\ f = \frac{d}{dx} g \right\} = \frac{d}{dx} A;$$

$$\widetilde{A} = A \Big/ \frac{d}{dx} A$$

$$f \longmapsto f \bmod \frac{d}{dx} A = \widetilde{f} \in \widetilde{A}$$

$$A \underset{\int \cdots dx}{\longrightarrow} \widetilde{A} \qquad \widetilde{f} = \int f\, dx$$

Problem.

$$f[u] \in \mathbb{C}[u_k^{(i)}];$$

$$\int_{-\infty}^{\infty} f[u(x)]\,dx = 0, \quad \forall u \in S(\mathbb{R}^1) \quad \Longleftrightarrow \quad f \in \frac{d}{dx}\mathbb{C}[u].$$

Def. 3. $A^{n-1} =$ space of tuples $(f_0, f_1, \ldots, f_{n-2})$, $f_k \in A$

[in the left margin: $\frac{\delta}{\delta u}$.]

$$\frac{\delta}{\delta u} : \widetilde{A} \longrightarrow A^{n-1}$$

$$\left(\frac{\delta \widetilde{f}}{\delta u}\right)_k = \sum_{\ell=0}^{\infty} \left(-\frac{d}{dx}\right)^{\ell} \frac{\partial f}{\partial u_k^{(\ell)}} = \frac{\delta \widetilde{f}}{\delta u_k}$$



Uncertain reading. The f in the numerator under the sum is heavily retraced.

Original source details: GELFAND-P000717

(here $\widetilde{f} = f \bmod \frac{d}{dx}A$)

GELFAND-P000718

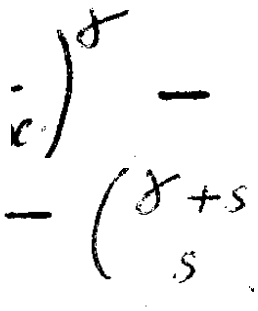
Def. 4. $\ell : A^{n-1} \longrightarrow A^{n-1}$

$$f = (f_0, \ldots, f_{n-2})$$

$$(\ell f)_r = \sum_s \ell_{rs} f_s$$

$$\ell_{rs} = \sum_{\gamma=0}^{n-1-r-s} \left[\binom{\gamma + r}{r} u_{r+s+\gamma+1} \left(-i\frac{d}{dx}\right)^{\gamma} - \right.$$

$$\left. - \binom{\gamma + s}{s} \left(i\frac{d}{dx}\right)^{\gamma} u_{r+s+\gamma+1} \right]$$



Uncertain reading. The manuscript repeats the minus at the line break, at the end of the first line and at the start of the second.

Original source details: GELFAND-P000718

$n = 2, 3$ easy; $n = 4$ — 1st substantial example.

Def. 5. $\forall f \in A^{n-1}$ we construct

$$\partial_f = \sum_{k,\ell} f_k^{(\ell)} \frac{\partial}{\partial u_k^{(\ell)}} \qquad \text{(a derivation)}$$

It is easy to check that

$$\partial_f \frac{d}{dx} = \frac{d}{dx} \partial_f$$

The operation is therefore defined $\partial_f : \widetilde{A} \longrightarrow \widetilde{A}$.

Def. 6. Let $f, g \in A^{n-1}$. Put

$$([f,g])_r = \partial_f g_r - \partial_g f_r$$

It is easy to check that

$$[\partial_f, \partial_g] = \partial_{[f,g]}.$$

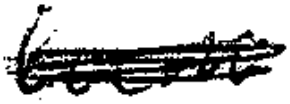
Hence the commutator $[\cdot,\cdot]$ defines on A^{n-1} the structure of a Lie algebra.

Def. 7. def. of the commutator in $\widetilde{A}$.

$$\{\widetilde{f}, \widetilde{g}\} = \int \sum_{r,s} \frac{\delta \widetilde{f}}{\delta u_r} \cdot \ell_{rs} \frac{\delta \widetilde{g}}{\delta u_s}\, dx$$

Theorem:

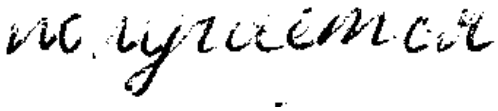
Undeciphered passage. A word after "Theorem" is heavily crossed out.



Original source details: GELFAND-P000718

we get a Lie algebra

Uncertain reading. The verb tentatively reads "получается" (we get), possibly "получится" (we will get).



Original source details: GELFAND-P000718

(i.e. the Jacobi identity is satisfied)

Problem: prove it more simply.

S. P. Novikov $A = \frac{d^3}{dx^3} + \left(u\frac{d}{dx} + \frac{d}{dx}u\right)$

also defines a Lie algebra.

(and the KdV eq. can be written with its help).

GELFAND-P000719

[source number: 56]

S. I. Gelfand. on the work of I. M. Gelf., I. N. Bernsh. and his own?:

Uncertain reading. The last word tentatively reads "свои." (his own). The final strokes run together.



Original source details: GELFAND-P000719

"The local structure of the category of Harish–Chandra modules"
or "Noncommutative algebraic geometry I".

G — a real Lie group, semisimple

$K \subset G$, K — a maximal compact subgroup

$\mathfrak{g}$ — the Lie algebra of the group G

Definition. A Harish–Chandra module M — is a linear space over $\mathbb{C}$ with actions of $\mathfrak{g}$ and K and with conditions (besides the natural conds. for the actions and compatibility of the actions of $\mathfrak{g}$ and K).

1) the actions of [crossed out: the Lie algebra of the group K] $\mathfrak{g}$ and K are compatible:

a) the Lie alg. of K acts on M as a subalgebra in $\mathfrak{g}$.

b) [crossed out: a)?]



Uncertain reading. A sign after "б)" is crossed out. It tentatively reads "а)".

Original source details: GELFAND-P000719

$$T\big(\underset{\uparrow}{k}\ X\ k^{-1}\big) = T_1(k)T(X)T_1(k^{-1}).$$
$$\begin{matrix} \uparrow & \uparrow \\ K & \mathfrak{g} \end{matrix}$$

T — the action of $\mathfrak{g}$, T_1 — the action of K

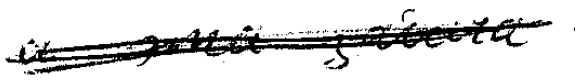
2) M decomposes with respect to K into a direct sum of [inserted above: irred.] representations and each [inserted below: irred.] repr. occurs a finite number of times.

H.–C. mods. form a category. Morphisms are defined in the natural way.

Problem: description of this category.

Idea: there is a noncommut. algebra responsible for this.

Gelfand, Ponomarev: for the Lorentz group [inserted below: $SL(2,\mathbb{C})$] it reduces to classifying pairs of matrices A, B, such that $AB = BA = 0$, [crossed out: and]



Undeciphered passage. Two more words after "and" are crossed out.

Original source details: GELFAND-P000719

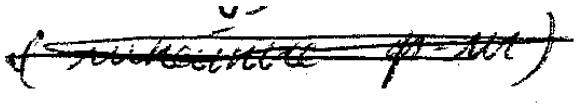
$U(\mathfrak{g})$ — the enveloping alg. of $\mathfrak{g}$

$Z(\mathfrak{g})$ — its centre

$$M \rightsquigarrow J_M = \{z \in Z(\mathfrak{g}) \mid zM = 0\} \qquad \text{(an ideal) in } Z(\mathfrak{g}).$$

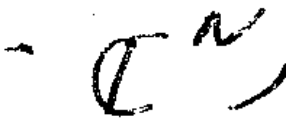
Θ — the set of characters $\theta : Z(\mathfrak{g}) \to \mathbb{C}$. (homomorphisms)

[crossed out beneath the formula: (linear formulas?)] $(\subset \mathbb{C}^N)$.



Uncertain reading. The crossed-out abbreviation after "linear" tentatively reads "ф-лы" (formulas).

Original source details: GELFAND-P000719



Uncertain reading. The case of the exponent in $\mathbb{C}^N$ is unclear, N or n.

Original source details: GELFAND-P000719

$$\operatorname{supp} M \subset \Theta$$
$$\operatorname{supp} M = \{\theta \mid J_M \subset \ker \theta\}$$

$\operatorname{supp} M$ — a finite set.

Let $D \subset \Theta$,

$$C_D = \{M \mid \operatorname{supp} M \subset D\}.$$

GELFAND-P000720

We can consid. H.–C. modules with supp. at a point $C_{\{\theta\}}$

Theorem (easy).

$$C_D = \bigoplus_{\theta \in D} C_{\{\theta\}}$$



Undeciphered passage. A mark between the equals sign and $\bigoplus$ is crossed out.

Original source details: GELFAND-P000720

in a certain sense,

(they decompose and there are no homomorphisms).

1st main theorem. Let D — a bounded domain in Θ (in the complex sense). Then $\exists$ $Z(\mathfrak{g})$-algebra Q



Uncertain reading. A small mark at the lower right of Q is crossed out. It may be a subscript.

Original source details: GELFAND-P000720

[crossed out: such] with a finite number a of additive $Z(\mathfrak{g})$-generators, [inserted above: such that] [crossed out: Then?] C_D is equivalent to the category of Q-modules that are finite-dimensional over $\mathbb{C}$.



Uncertain reading. The crossed-out word before C_D tentatively reads “Тогда” (Then).

Original source details: GELFAND-P000720

(∃ a functor realizing the equivalence)

Q — $Z(\mathfrak{g})$

— · —

"Izvestia" and "Socialist Industry" [crossed out: dated the 23rd]

$23/\overline{\text{IX}}\text{-}77$ $22/\overline{\text{IX}}\text{-}77$

a report from Petrozavodsk of an unidentified flying object

— · —

Example.

$$G = SL_2(\mathbb{R})/\{\pm 1\}$$
$$Z(\mathfrak{g}) \cong \mathbb{C}[z]$$
$$\Theta = \mathbb{C}^1$$

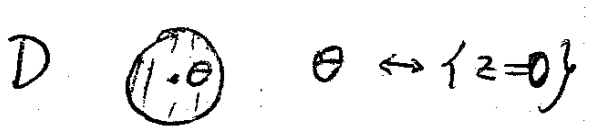


Diagram and typed description.

D  $\theta \leftrightarrow \{z = 0\}$

Original source details: GELFAND-P000720

We will construct the algebra over analytic fns.

$$Q = (a_{ij}(z))_{i,j=1}^{3} \text{ — matr. } 3 \times 3, \qquad a_{ij}(z) \text{ — entire fns.}$$

$$(a_{ij}(0)) = \begin{pmatrix} * & 0 & 0 \\ * & * & 0 \\ * & 0 & * \end{pmatrix}$$

GELFAND-P000721

10/X—77 Gelfand Sem..

S. Gelfand.

G — a semisimple real Lie group

K — a maximal compact subgroup

$\mathfrak{g}$ — the Lie algebra of the group G

$U(\mathfrak{g})$ — the enveloping algebra.

$Z(\mathfrak{g})$ — the centre of $U(\mathfrak{g})$

Harish–Chandra module (HCM) V, if

[H.–C. thm.: in an irreduc. repr. of a semisimple group, every [inserted above: irred.] repr. of the max. comp. group K occ. a finite no. of times]

In V there are given reprs. T_1 of the alg. $U(\mathfrak{g})$, T_2 — a repr. of the group K.

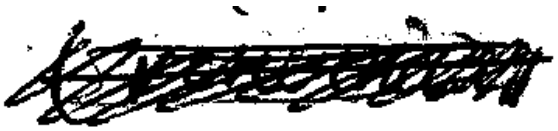
1) V — a finitely gen. $U(\mathfrak{g})$-module.

2) $T_1|_{\mathfrak{k}} = dT_2$,

$$T_2(k)T_1(X)T_2(k^{-1}) = T_1(kXk^{-1})$$

3) $T_2 = \bigoplus$ irred. cont. reprs. of K, and each occurs a finite number of times.

(direct sum in the literal sense).

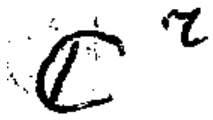


Undeciphered passage. A short passage before "Result" is heavily crossed out.

Original source details: GELFAND-P000721

Result: let

$$\Theta = \{\theta : Z(\mathfrak{g}) \to \mathbb{C}\} = \mathbb{C}^r$$



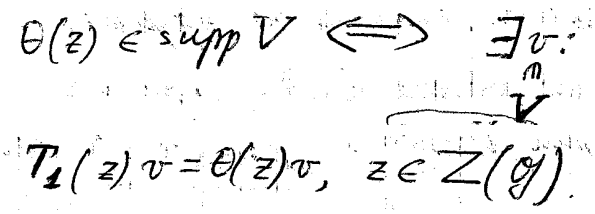
Uncertain reading. The exponent tentatively reads r, possibly τ.

Original source details: GELFAND-P000721

(ring homomorphisms)

$$V \mapsto \operatorname{supp} V \subset \Theta$$

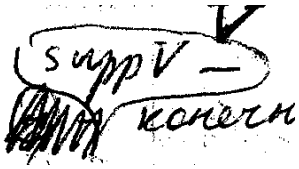
$$\theta(z) \in \operatorname{supp} V \iff \exists \underset{V}{v} : T_1(z)v = \theta(z)v, \quad z \in Z(\mathfrak{g}).$$



Uncertain reading. The manuscript has $\theta(z)$ on the left and gives no explicit condition $v \neq 0$.

Original source details: GELFAND-P000721

In fact



Original source details: GELFAND-P000721

Undeciphered passage. A short passage under the insertion “supp V —” is heavily crossed out.

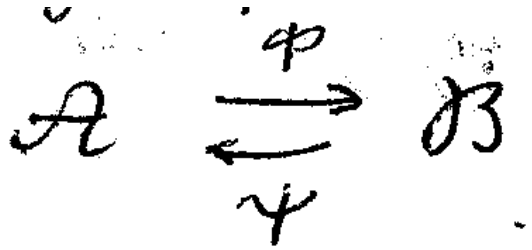
[inserted above: supp V —] a finite set.

Let there be a bounded domain $D \subset \Theta$,

$$C_D = \{V : \operatorname{supp} V \subset D\}.$$

For disjoint sets D_1 and D_2 we get the direct sum of categories.

Equivalence of categories:



Original source details: GELFAND-P000721

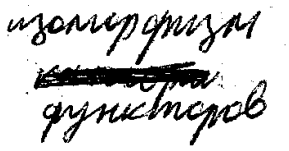
Diagram and typed description.

$$\mathcal{A} \underset{\psi}{\overset{\varphi}{\rightleftarrows}} \mathcal{B}$$

φ, ψ — functors

$$\varphi \circ \psi \underset{\uparrow}{\sim} \mathrm{id}_B, \qquad \psi \circ \varphi \underset{\nearrow}{\sim} \mathrm{id}_A$$

[arrows from the inscription to both signs] isomorphism



Original source details: GELFAND-P000721

Undeciphered passage. A word between “isomorphism” and “of functors” is heavily crossed out.

of functors

GELFAND-P000722

[source number: 61]

Isomorphism of functors:

$$\forall a: \qquad \psi\varphi(a) \longleftrightarrow a$$

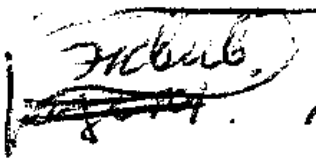
(do not coincide, but are isomorphic!) and all diags. commute.

Example of equiv. but non-isom. categories:

1) categ. of finite-dimensional lin. spaces over $\mathbb{C}$

2) — ″ — of finite-dimensional modules over $M(2, \mathbb{C})$.

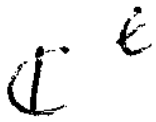
The category of HCMs [enclosed insertion above: is equiv. to] a certain category of finite-dimensional modules.



Original source details: GELFAND-P000722

Undeciphered passage. A word below the insertion “эквив.” (equiv.) is heavily crossed out.

$\underset{\text{open}}{E} \subset \mathbb{C}^{\ell}$, B — a direct sum of several matrix algebras.



Original source details: GELFAND-P000722

Uncertain reading. The exponent on $\mathbb{C}$ tentatively reads ℓ.

$B(E)$ — analytic fns. on E with values in B

Def. A subalg. $R \subset B(E)$ is called nice, if

1) it cont. all scalar fns.:

$$f(z)\mathbf{1}_B \in R$$

2) $\{f(z), f \in R\} = B$ for almost all $z \in E$

(i.e. for all except an analytic subset of smaller dimension)

[crossed out: Theorem?]



Original source details: GELFAND-P000722

Uncertain reading. The crossed-out heading tentatively reads “Теорема” (Theorem).

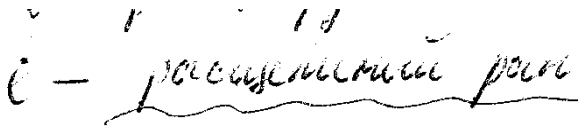
Mod R — categ. of R-modules that are finite-dimensional lin. spaces.

Theorem. $C_D \sim \text{Mod}\, R$

for a certain nice alg. R.

r — rank of the group

ℓ — split rank of the group.



Uncertain reading. The sign before "split rank" tentatively reads ℓ.

Original source details: GELFAND-P000722

— · —

The rest will be on an example.

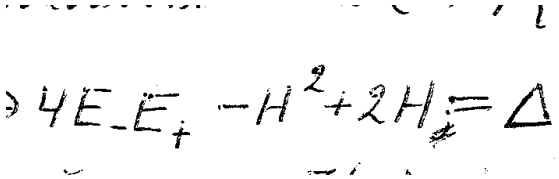
$$G = [\text{crossed out } SL(2,\mathbb{R})?] \quad SL_2(\mathbb{R})/\{\pm 1\}, \qquad \underset{\text{Lie alg.}}{\mathfrak{sl}_2(\mathbb{R})} = \{E_+, E_-, H\}$$



Uncertain reading. The crossed-out expression after $G =$ tentatively reads $SL(2,\mathbb{R})$.

Original source details: GELFAND-P000722

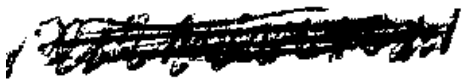
$$Z(\mathfrak{g}) \ni 4E_-E_+ - H^2 + 2H := \Delta \qquad E_+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad E_- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$



Uncertain reading. A minus is visible before H^2. Small crossed-out strokes lie below the defining sign for Δ.

Original source details: GELFAND-P000722

Every other element of $Z(\mathfrak{g})$ is a polynomial



Undeciphered passage. A passage after "polynomial" is heavily crossed out.

Original source details: GELFAND-P000722

in Δ.

$\Theta = \mathbb{C}^1$, $D =$ a neighbourhood of the point 0, suff. small



Uncertain reading. The sign after "point" is open at the right and tentatively reads 0.

Original source details: GELFAND-P000722

A nice subalgebra:

$$E = D, \qquad B = M_3(\mathbb{C}), \qquad R \subset \begin{pmatrix} a_{11}(z) & a_{12}(z) & a_{13}(z) \\ \cdot & - & - \\ \cdot & - & - \end{pmatrix} = (a_{ij}(z))$$

GELFAND-P000723

The condition singling out R:

$$a_{ij}(0) = \begin{pmatrix} * & 0 & 0 \\ * & * & 0 \\ * & 0 & * \end{pmatrix}$$

There is a basis of 9 elements over the fns.

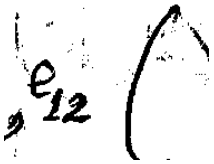
$$e_{11} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad e_{12} \begin{pmatrix} 0 & z & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad e_{21} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

and so on.



Original source details: GELFAND-P000723

Uncertain reading. The zero in the first entry of the matrix by e_{12} is heavily retraced.



Original source details: GELFAND-P000723

Uncertain reading. No equals sign is visible between e_{12} and the opening parenthesis.

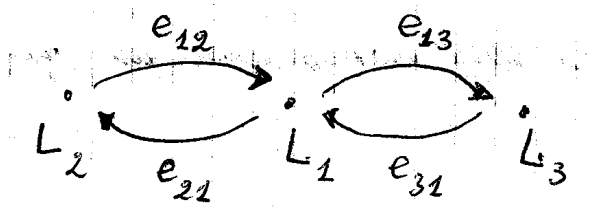
$$e_{21}e_{12} = ze_{22}$$

We want to find finite-dimensional modules for which mult. by z is a nilpot. operator.

$$e_{ii}^2 = e_{ii}.$$

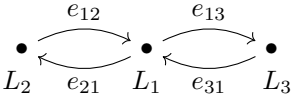
$$L_i = \{m : \ e_{ii}m = m\}.$$

$$M = L_1 \oplus L_2 \oplus L_3$$



Original source details: GELFAND-P000723

Diagram and typed description.



In the manuscript, e_{13} points from L_1 to L_3, and e_{31} from L_3 to L_1.

$e_{21}e_{12} = ze_{22}$ is nilpotent

$$e_{12}e_{21} = ze_{11} = e_{13}e_{31}$$

hence the categ. of modules over this ring $\cong \ldots$

For $SL(3,\mathbb{R})$ the answer is as follows:

$$E = \{z_1, z_2\} \qquad \text{in a nbhd. of zero:}$$

$$B = \underset{a_{ij}}{M_4} \oplus \underset{b_{ij}}{M_3} \oplus \underset{c_{ij}}{M_3} \oplus \underset{d}{M_1}$$

Conditions.

1) At $z_1 = 0$ $\quad \Big\| \quad a_{13} = a_{14} = a_{23} = a_{24} = b_{12} = b_{13} = 0$, $a_{33} = b_{22}, \quad a_{34} = b_{23}, \quad a_{43} = b_{32}, \quad a_{44} = b_{33}$

GELFAND-P000724

[source number: 62]

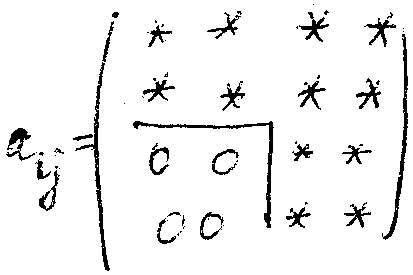


Diagram and typed description.

$$a_{ij} = \left(\begin{array}{cccc} * & * & * & * \\ * & * & * & * \\ \hline 0 & 0 & * & * \\ 0 & 0 & * & * \end{array}\right)$$

Original source details: GELFAND-P000724

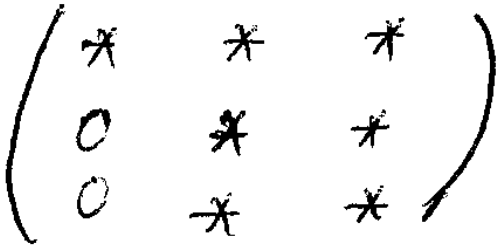
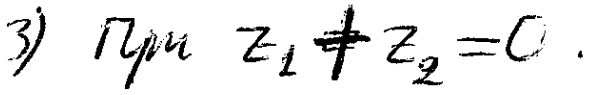


Diagram and typed description. $\begin{pmatrix} * & * & * \\ 0 & * & * \\ 0 & * & * \end{pmatrix}$

Original source details: GELFAND-P000724

2) When $z_2 = 0$ $\quad \Big\| \quad a_{12} = a_{14} = a_{34} = c_{12} = c_{13} = a_{32} = 0$, $a_{22} = c_{22}, \quad a_{24} = c_{23}, \quad a_{42} = c_{32}, \quad a_{44} = c_{33}$

3) When $z_1 + z_2 = 0$. $\quad \Big\| \quad b_{13} = b_{23} = c_{13} = c_{23} = 0;$ $b_{33} = c_{33}.$



Uncertain reading. The overwritten sign between z_1 and z_2 is tentatively read as $+$.

Original source details: GELFAND-P000724

GELFAND-P000725

24/X–77

S. I. Gelfand. Harish-Chandra Modules

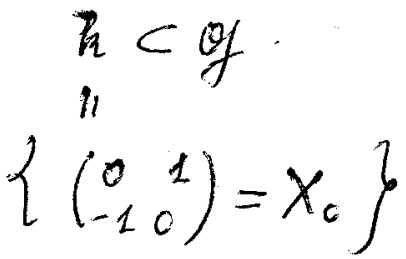
HC modules for

$$G = SL(2,\mathbb{R})/(\pm 1)$$
$$K = SO_2/(\pm 1)$$
(max. comp. subgroup)

$\mathfrak{g}$ — Lie algebra

$$\underset{\begin{pmatrix}0 & 1\\ 0 & 0\end{pmatrix}}{E_+}, \quad \underset{\begin{pmatrix}0 & 0\\ 1 & 0\end{pmatrix}}{E_-}, \quad \underset{\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}}{H}$$ — generators of $\mathfrak{g}$

$$\begin{array}{l}\mathfrak{k} \subset \mathfrak{g}.\\ \|\\ \left\{\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix} = X_c\right\}\end{array}$$



Uncertain reading. The subscript on X reads as an open c.

Original source details: GELFAND-P000725

GELFAND-P000726

[source number: 67]

A module H.–C. — is a (infinite-dimensional) repr. in [crossed out] V, such that



Uncertain reading. The crossed-out fragment before V is illegible.

Original source details: GELFAND-P000726

1) X_0 has only integer eigenvalues, and even ones at that

2) [crossed out] each eigenvalue [inserted above: of the operator X_0] has finite multiplicity



Uncertain reading. The heavily crossed-out mark after 2) is illegible.

Original source details: GELFAND-P000726

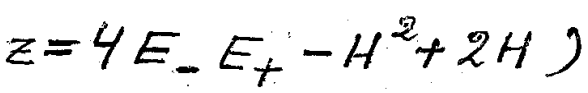
3) V is finitely generated (over the envelop. algebra)

($\Leftrightarrow$ the eigenvalue multiplicities are bounded).

$v_1, \dots, v_n$ — generators

> *[In the right margin beyond the double line]* 1) — so that one can integrate to a group repr. (e^{inx} etc.)

Definition. $Z(\mathfrak{g})$ — the centre of the envelop. algebra $U(\mathfrak{g})$
(gen. by the element $z = 4E_-E_+ - H^2 + 2H$)



Uncertain reading. The manuscript has a minus before H^2 and a plus before $2H$.

Original source details: GELFAND-P000726

supp $V =$ the set of eigenvalues of the operator z in V

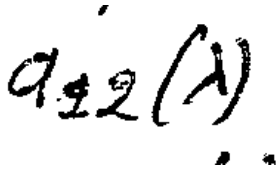
Theorem: this is a finite set for any V.

Definition. Let D — a domain in $\mathbb{C}$.

$$C_D = \{V,\ \text{supp}\, V \subset D\}$$

Let D — a small neighbourhood of zero (disc of radius $< 1/2$).

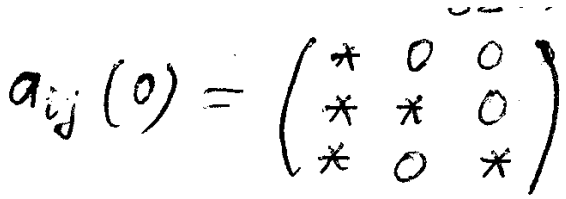
$$R = \left\{ \begin{pmatrix} a_{11}(\lambda) & a_{12}(\lambda) & a_{13}(\lambda) \\ a_{21}(\lambda) & a_{22}(\lambda) & a_{23}(\lambda) \\ a_{31}(\lambda) & a_{32}(\lambda) & a_{33}(\lambda) \end{pmatrix} \right. , \quad a_{ij}(\lambda) \text{ — analytic fns. in } D$$



Uncertain reading. The first subscript on $a_{12}(\lambda)$ is overwritten and reads as 1.

Original source details: GELFAND-P000726

$$a_{ij}(0) = \begin{pmatrix} * & 0 & 0 \\ * & * & 0 \\ * & 0 & * \end{pmatrix}$$



Uncertain reading. The left-hand side is written $a_{ij}(0)$.

Original source details: GELFAND-P000726

$$\operatorname{Mod} R = \left\{ \underset{\substack{| \\ R\text{-module}}}{M} \ : \ \dim_{\mathbb{C}} M < \infty \right\}$$

Theorem. The category C_D is equivalent to the category Mod R.

Example.

The category of modules over the matrix ring is equivalent to the category of [crossed out: linear...?] finite-dimensional linear spaces



Original source details: GELFAND-P000726

Uncertain reading. The crossed-out word tentatively begins “linear”; the ending is uncertain.

($\sim$ a repr. of the matrix alg. is a direct sum)

V — the stand. repr. of the alg. M_ℓ



Original source details: GELFAND-P000726

Uncertain reading. The subscript on M is tentatively read as ℓ; the mark also resembles e.

$$E \longmapsto V \underset{\mathbb{C}}{\otimes} E, \qquad M_\ell \text{ acts on the first variable}$$

$$W \text{ — } M_\ell\text{-module} \quad \longmapsto \quad E = \operatorname{Hom}_{M_\ell}(V, W).$$

(the 2 functors giving the equivalence are written out)

Old formulation: $\forall\, M_\ell$ module decomposes as a direct sum.

GELFAND-P000727

Proof of the theorem.

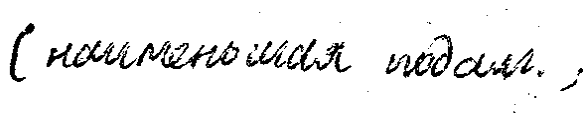
I. Let a list of numbers be given $n_1, \dots, n_k$. Construct from it the ring $Q_{n_1,\dots,n_k}$

a) $J_{n_1 \dots n_k}$ — the left ideal in $U(\mathfrak{g})$, generated by the element

$$(X_0 - 2in_1)(X_0 - 2in_2) \cdots (X_0 - 2in_k)$$

b) $\quad N_{n_1,\dots,n_k} = \{u \in U(\mathfrak{g}) \mid J_{n_1,\dots,n_k} u \subset J_{n_1,\dots,n_k}\}$

(the smallest subalg., in which J — a 2-sided ideal



Original source details: GELFAND-P000727

Uncertain reading. The manuscript reads “smallest”.

$$Q_{n_1 \dots n_k} = N_{n_1,\dots,n_k}/J_{n_1,\dots,n_k}$$

Def. V — an H.–C. module

$$\tau_{n_1,\dots,n_k}(V) \underset{\text{def}}{=} \{v : \ (X_0 - 2in_1)\cdots(X_0 - 2in_k)v = 0\}$$

This is a finite-dimensional subspace

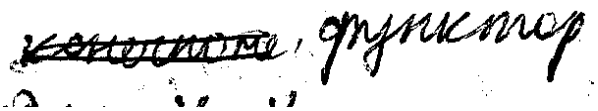
Dimension $=$ sum of the multiplicities

N maps it into itself

J —"— to zero

$$\Rightarrow \quad Q_{n_1,\dots,n_k} \text{ acts in } \tau_{n_1,\dots,n_k}(V)$$

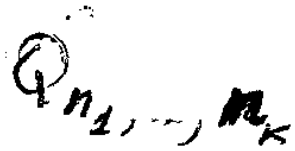
This is [crossed out: construction?], a functor



Original source details: GELFAND-P000727

Uncertain reading. The crossed-out word is tentatively read as "construction".

H.–C. modules $\longrightarrow$ finite-dimensional modules over $Q_{n_1,\dots,n_k}$



Original source details: GELFAND-P000727

Uncertain reading. The first mark in the last subscript of Q is retraced. It reads n_k.

It is clear that

$$z \in N_{n_1,\dots,n_k}$$

$$\text{supp } M = \{\text{eigenvalues of } z\}$$

$$(\text{Mod } Q_{n_1,\dots,n_k})_D$$

(finite-dimensional modules with support in D).



Original source details: GELFAND-P000727

Diagram and typed description.

$\longmapsto$

The short arrow points left and has a crossbar at the right. Its connection to the adjacent line is uncertain.

Theorem 1. Let D — be any bounded domain in $\mathbb{C}^1$. Then there exist $n_1, \dots, n_k$, such that C_D and $(\text{Mod } Q_{n_1,\dots,n_k})_D$ are equiv.

GELFAND-P000728

[source number: 68]

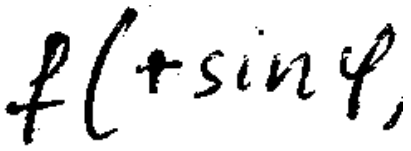
II. Definition of the reprs. of the principal series.

from s construct the module $T^{(s)} = V$

$$V = \{ \underset{\substack{\text{functions} \\ \text{on the plane}}}{f(x,y)} \}$$

1) $f(tx, ty) = |t|^s f(x, y)$

2) $f(+\sin\varphi, \cos\varphi) = P(\sin\varphi, \cos\varphi)$



Uncertain reading. The first argument has a $+$ before $\sin\varphi$.

Original source details: GELFAND-P000728

P — a polynomial.

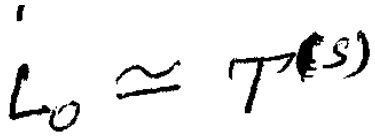
The Lie algebra acts in the natural way (diff. operators)

(the group acts on the argument)

$$E_+ \sim x\frac{\partial}{\partial y}; \quad E_- \sim y\frac{\partial}{\partial x}; \quad H \sim x\frac{\partial}{\partial x} - y\frac{\partial}{\partial y}.$$

Restr. to $S^1 \cong$ the space of polynomials (even on the circle)

$$L_0 \simeq T^{(s)} \qquad \text{(for any } s\text{)}.$$



Uncertain reading. The opening parenthesis in the superscript (s) is heavily retraced.

Original source details: GELFAND-P000728

Lemma 1. Let $X \in U(\mathfrak{g})$. The action of X depends polynomially on s.

(Matrix entries in the monomial basis are polynomials in s)

$$\tau_{n_1,\ldots,n_k}(T^{(s)}) = L_{n_1,\ldots,n_k}$$

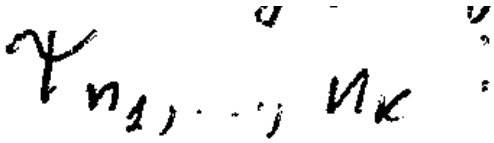
Lemma 2. Let $q \in Q_{n_1,\ldots,n_k}$. The action of q depends polynomially on s.

Theorem 2. Set

$$B_{n_1,\ldots,n_k} = \mathrm{End}(L_{n_1,\ldots,n_k})$$

Then there exists a map

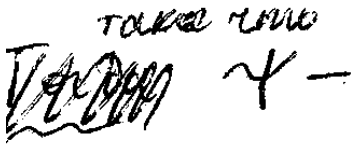
$$\psi_{n_1,\ldots,n_k} : Q_{n_1,\ldots,n_k} \longrightarrow B_{n_1,\ldots,n_k}[s],$$



Uncertain reading. The symbol for the map is tentatively read as ψ.

Original source details: GELFAND-P000728

[crossed out: Thm.?] [inserted above: such that] ψ — an embedding



Original source details: GELFAND-P000728

Uncertain reading. The crossed-out heading tentatively reads “Teop.” (Thm.).

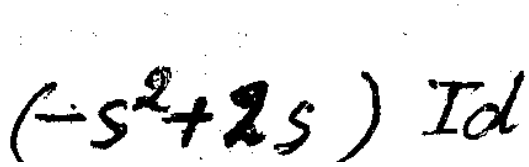
(i.e. the principal series contains as sub- and quotient reprs. $\forall$ HC modules)

III. Combining stages I and II.

D — a small nbhd. of zero

Lemma 0.

On $T^{(s)}$ the operator z is $(-s^2+2s)\,\mathrm{Id}$



Original source details: GELFAND-P000728

Uncertain reading. The manuscript gives $(-s^2+2s)$ Id.

$z=0$ for $s=0$ and $s=2$.

Lemma 1. Let s be near the point 0.

1) Then T^s, $s\neq 0$ is irreducible

$$0 \longrightarrow \mathrm{Id} \longrightarrow T^{(0)} \longrightarrow T_+ \oplus T_- \longrightarrow 0$$



Original source details: GELFAND-P000728

Uncertain reading. The second letter of Id is heavily retraced.

and the exact seq. does not split

GELFAND-P000729

$$2)\quad T^{(s)}\big|_K = \bigoplus_{j=-\infty}^{\infty} \delta_j.$$

$\delta_k =$ eigensp. of X_0 with eigenval. $2ik$.

(1-dimensional repr.)

$$I\big|_K = \delta_0, \qquad T_+\big|_K = \bigoplus_{j>0} \delta_j, \qquad T_-\big|_K = \bigoplus_{j<0} \delta_j.$$



Original source details: GELFAND-P000729

Uncertain reading. A narrow mark resembling I is visible beyond the right border. Its connection to the text is uncertain.

[In the left margin: B?] Lemma 2. Near $s = 2$



Uncertain reading. The heavily retraced mark in the margin tentatively reads “B”.

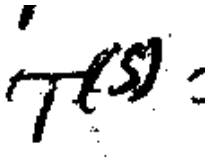
Original source details: GELFAND-P000729

$$0 \longrightarrow T_+ \oplus T_- \longrightarrow T^{(2)} \longrightarrow I \longrightarrow 0$$

$T^{(s)}$, $s \neq 2$ is irreducible

Lemma 3. $s \neq 0$ $\quad T^{(s)} \simeq T^{(2-s)}$

s near zero



Uncertain reading. The opening parenthesis in the superscript of $T^{(s)}$ is retraced.

Original source details: GELFAND-P000729

Consider the numbers $(n_1, n_2, n_3) = (-1, 0, 1)$ and the ring

$$Q_{(-1,0,1)}$$

D — a suff. small nbhd. of zero $\quad \Rightarrow \quad C_D \sim Q_{(-1,0,1)}$

(explanation: in T_+ there is a repr. with $+1$, in T_- — ” — with -1 etc.)

$$\tau_{-1,0,1}(T^{(s)}) = \{e_{-1}, e_0, e_1\}$$

$q \in Q_{-1,0,1}$

. . . . . .

Done: $SL(2,\mathbb{R})/(\pm 1)$, $SL(2,\mathbb{R})$, $PGL(2,\mathbb{R})$,
$SL(2,\mathbb{C})$, $SL(3,\mathbb{R})$

Probably possible $SL(3,\mathbb{C})$ and $SU(2,1)$

Recommended, calc. $SO(4,1)$ (easy), $Sp(4,\mathbb{C})$ (easy)

Difficult: $SL(4,\mathbb{R})/(\pm 1)$, $SL(4,\mathbb{C})$.

Read the works of Hirai, Zhelobenko

Gelfand, Graev

Krichever

2 minutes for a sensational announcement.

[In the left margin: On eqns.?]

$$[L_1, L_2] = 0$$



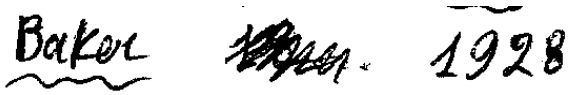
Uncertain reading. The Russian abbreviation tentatively reads “Об ур-и”, possibly “Об ур-й”.

Original source details: GELFAND-P000729

Coddington pointed to the works of

Burchnal, Chaundy 1922, 1928, 1931

and Baker [crossed out] 1928



Uncertain reading. The writing between “Baker” and “1928” is heavily crossed out.

Original source details: GELFAND-P000729

[To the right of the list of works:] — they objected to Baker, who recommended using alg. geom.

The problem of describing commuting ops. with scalar coeffs. of relatively prime ord. was posed and solved.

the answer uses no Θ-fns. and is insufficiently effective

the 1st was found via Lamé fns. (as in Akhiezer). they did not see the straight-line behav.

GELFAND-P000730

[source number: 69]

path on the Jacobian (and even conditional periodicity).

Forgotten — in the 1939 paper there are no longer any references.

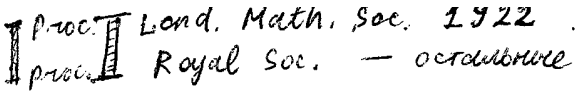


Diagram and typed description.
Proc. Lond. Math. Soc. 1922.
Proc. Royal Soc. — the others

Original source details: GELFAND-P000730

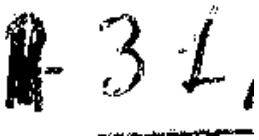
almost commuting [crossed out] and Lax pairs exist.



Uncertain reading. A small mark after “almost commuting” is crossed out.

Original source details: GELFAND-P000730

[crossed out] 31/X–77 Gelfand Sem. S. P. Novikov. Instantons.



Uncertain reading. The heavily crossed-out mark before the date is illegible.

Original source details: GELFAND-P000730

$$\mathbb{R}^4 \subset S^4 \qquad (|x| \to \infty)$$

$A =$ [crossed out] $A_\mu \, dx^\mu$ connection form



Uncertain reading. The crossed-out expression after $A =$ appears to contain A and d. The indices are illegible.

Original source details: GELFAND-P000730

If the curvature is 0, then

$$(1) \qquad A_\mu \longrightarrow \frac{\partial g}{\partial x^\mu}\, g^{-1}(x), \qquad g(x) \in G$$

$$A_\mu(x) \in L_G \qquad \text{(Lie algebra)}$$

Curvature form:

$$(2) \qquad F_{\nu\mu} = dA + [A, A] = \mathcal{D}A$$

(covariant derivative)

Most often $G = SU_2$ (Yang–Mills).

$[A, A]$ — exterior prod., but mult. of coeffs. is in the Lie alg..

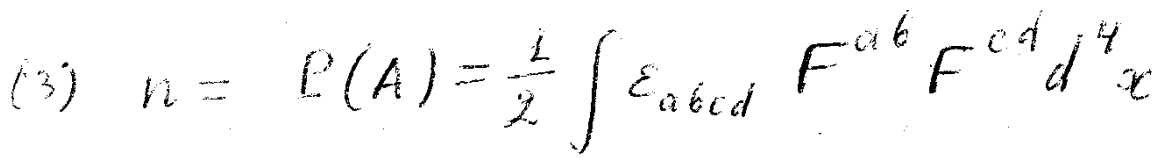
Other notation: $A \wedge A = [A, A]$, $\mathcal{D}A = \nabla_A A$

The requirement that everything be on S^4 — is slightly stronger than finite energy.

(Rem. Nonlinear (chiral) fields $\leftrightarrow$ connection with zero curvature; given by fns. with values in the group.)

Set

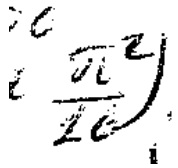
$$(3) \qquad n = P(A) = \frac{1}{2}\int \varepsilon_{abcd} F^{ab} F^{cd}\, d^4x$$



Original source details: GELFAND-P000730

Uncertain reading. Equation (3) has no written trace symbol before the product $F^{ab}F^{cd}$.

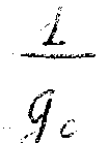
(char. Chern class — top. invariant — an integer up to a factor $\frac{\pi^2}{16}$).



Original source details: GELFAND-P000730

Uncertain reading. The denominator of the fraction is tentatively read as 16.

$$(4) \qquad S(A) = \frac{1}{g_c}\int_{\mathbb{R}^4} \mathrm{Sp}(F_{\mu\nu}F^{\mu\nu})\, d^4x$$

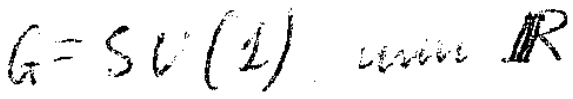


Original source details: GELFAND-P000730

Uncertain reading. The open subscript on g tentatively reads c, possibly 0.

(Yang–Mills functional)

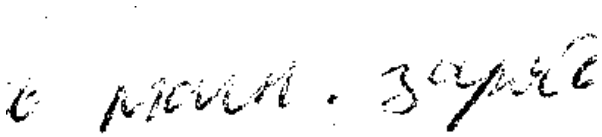
Class. electrodynamics — $G = SU(1)$ or $\mathbb{R}$



Original source details: GELFAND-P000730

Uncertain reading. The manuscript reads $SU(1)$.

depending on whether there are real charges or not.



Uncertain reading. The faint abbreviation before “charges” tentatively reads “реал.” (real).

Original source details: GELFAND-P000730

Yang–Mills eqns.:

$$(5) \qquad \delta S = 0.$$

Class. problem: find all such A, that $S(A) < +\infty$.

Rem. The metric of S^4 is conformally equiv. to the metric of $\mathbb{R}^4$,

$$g_{\mu\nu}(y) = g(y)\delta_{\mu\nu}$$

The Maxwell and Yang–Mills eqns. are conformally invariant.

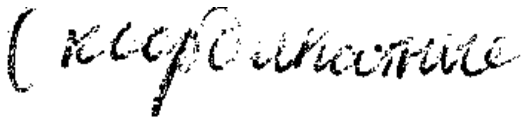
Grp. of conformal transfs. of Euclidean sp. and the sphere $\quad \| \quad O(5,1) = K_4$

GELFAND-P000731

Solutions are mult. by a scalar under the action of the conformal group:

$$f : X \longrightarrow X, \qquad f^*(g_{ij}) = g(x)g_{ij}.$$

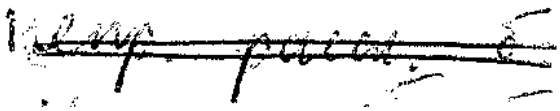
(continuous conf. transfs.)



Uncertain reading. The first word in parentheses tentatively reads “continuous”.

Original source details: GELFAND-P000731

[crossed out: Cont. bund. into?]



Uncertain reading. The crossed-out line tentatively reads “Непр. рассл. в” (Cont. bund. into).

Original source details: GELFAND-P000731

A cont. map of bases takes a bundle to a bundle and a connection to a connection. Then the new connection again satisfies the Yang–Mills eqn..

$O(5,1)$

On the sphere S^4 there is $O(5)$

Stereogr. transf. S^4 onto $\mathbb{R}^4$ and in $\mathbb{R}^4$ take similarity transfs..

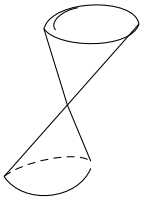
Where does $O(5,1)$?

We must consider the cone

$$x_1^2 + x_2^2 + \cdots + x_5^2 - x_6^2 = 0.$$



Diagram and typed description.



Original source details: GELFAND-P000731

[undeciphered mark] and lin. transfs. preserving the cone.

Uncertain reading. The isolated mark before "and" is illegible.



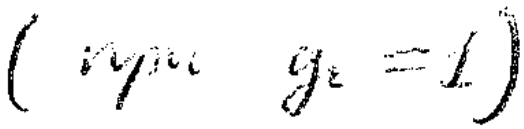
Original source details: GELFAND-P000731

The rays of the cone — S^4. The transfs. are conformal on S^4.

The conformal group preserves the functional $S(A)$

Belavin–Polyakov instantons.

$$(6) \qquad \min_{P(A)=n} S(A) = n. \qquad \text{(when } g_c = 1\text{)}$$



Uncertain reading. The subscript on g is tentatively read as c, possibly 0.

Original source details: GELFAND-P000731

and in fact the minima satisfy the eqn. [crossed out]



Uncertain reading. The crossed-out expression resembles $F = *F$; the middle signs are uncertain.

Original source details: GELFAND-P000731

$$(7) \qquad \underline{F = *F}$$

(duality eqns.)

(Belavin and Polyakov)

And conversely, solns. of these eqns. with given n — realize a strict minimum in the given class.

A number of authors gave explicit solns. (for $\forall\, n$ at least one solution exists).

[crossed out: Ger?] t' Hooft found explicit fmlas. for $\underline{n = 2}$. (all solutions!)



Uncertain reading. The crossed-out fragment before the surname tentatively reads "Гер" (Ger).

Original source details: GELFAND-P000731

by means of an elem. substitution.

The solutions form an $(8n-3)$-dimensional manifold (Schwarz, from the Atiyah–Singer thm.). This is M^{8n-3}.

Atiyah–Singer 3 months later.

t' Hooft found [crossed out] a manifold of dimension $\underline{\underline{5n+4}}$. ($n \geq 3$)

Uncertain reading. The short crossed-out entry after "found" is illegible.



Original source details: GELFAND-P000731

For $n = 2$ they coincide. It is unclear whether there is another connected component.

Proof of Schwarz's thm.

Linearize eqns. (7):

$$(A_\mu) \longrightarrow A_\mu + a_\mu = A + \delta A$$

linearized operator $\quad a_\mu \longrightarrow Ta_\mu$



Uncertain reading. An isolated, undeciphered trace is visible to the right of Ta_μ.

Original source details: GELFAND-P000731

GELFAND-P000732

[source number: 70]

$$B(A) = F - *F = 0,$$
$$B(A + \delta A) = T(\delta A) + O((\delta A)^2)$$

Counting the index: cn — a linear fn. of n; compute the vals. at 0 and 1.

Cokernel $= 0$ [crossed out] $\Leftarrow$ Hopf considerations.



Uncertain reading. A passage after "Cokernel = 0" is heavily crossed out. A left-pointing arrow is visible at right.

Original source details: GELFAND-P000732

Obtain $8n$: 3 parameters redundant due to gauge invariance.

Use of the conf. group $\Rightarrow$ For $n = 1$ all solns. are spher. symmetric.

$O(5,1)$ acts on the set of solns.

For $n = 1$ the set of solns. is 5-dim. [At right:] $\dim O(5,1) = 15$.

on $\forall$ conn. comp. acts $O(5,1)$.

Hence the stab. subgroup has dim. ≥ 10. But > 10 is impossible, since the stab. subgroup is compact (there is a metric)

$\Rightarrow$ they are homog. spaces for $SO(5)$, i.e. spher. symmetric.

(spher. symm. up to gauge invariance).

2-instant. — still incomplete.

Method of Atiyah and Zakharov–Belavin

Cond. $F = *F$.

we want to express it as a cond. of commutativity of diff. operators

$$\left(\text{e.g., } \left[L - \frac{\partial}{\partial t},\ A - \frac{\partial}{\partial y}\right] = 0\right)$$

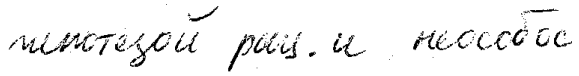
Theorem

$$(8) \quad [\widehat{B}_1(\lambda), \widehat{B}_2(\lambda)] = 0,\ \forall\lambda \quad \Longleftrightarrow \quad \text{duality eqns. (7).}$$

(1st order in λ).

[At right, ‖] (Zakharov, Belavin, Yang).

Single out solns. by assuming rationality and nonsingularity.



Uncertain reading. The words after "Single out solns." tentatively read "гипотезой рац." (by assuming rationality).

Original source details: GELFAND-P000732

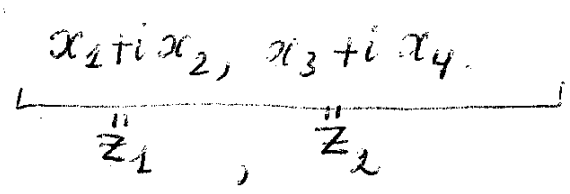


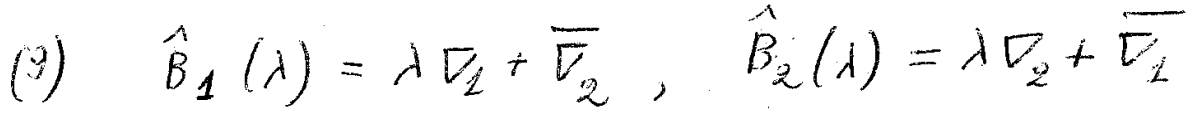
Diagram and typed description.

$$\underbrace{x_1 + ix_2,\quad x_3 + ix_4}_{\begin{matrix}\| & \| \\ z_1 & z_2\end{matrix}}$$

Original source details: GELFAND-P000732

$B_j(\lambda)$ themselves write as covar. diff. operators.

$$(9)\quad \widehat{B}_1(\lambda) = \lambda\nabla_1 + \overline{\nabla}_2, \qquad \widehat{B}_2(\lambda) = \lambda\nabla_2 + \overline{\nabla}_1.$$



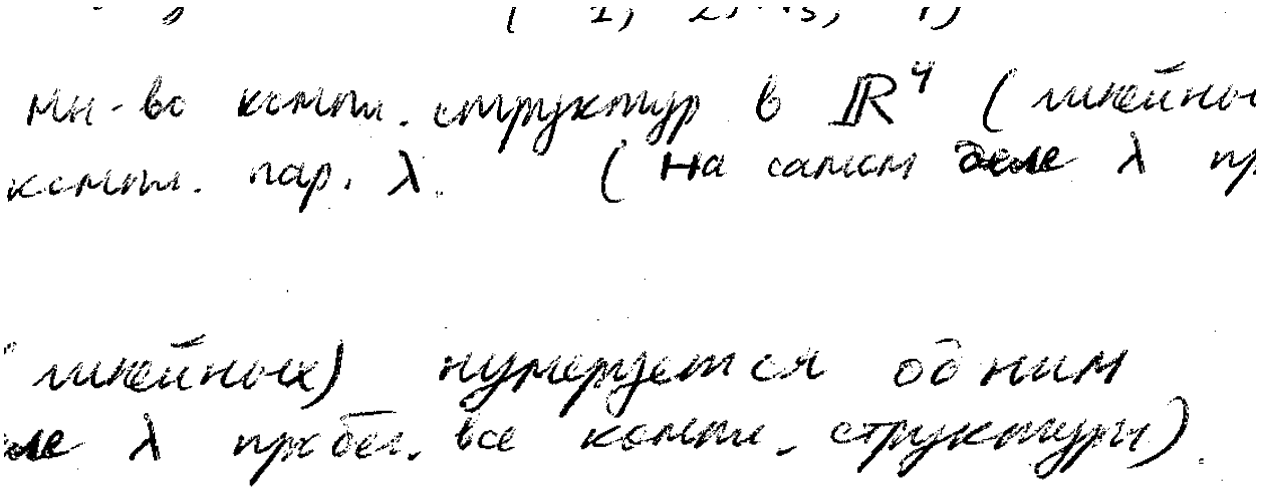
Original source details: GELFAND-P000732

Uncertain reading. Both operators in (9) have a written $+$ before the conjugate derivative.

$$(10)\quad \nabla_1 = A_1 + iA_2 + \frac{\partial}{\partial z_1}, \qquad \nabla_2 = A_3 + iA_4 + \frac{\partial}{\partial z_2}.$$

Connection $= \{A_1, A_2, A_3, A_4\}$.

The set of complex structures in $\mathbb{R}^4$ (linear) is indexed by one complex param. λ. (In fact λ runs through all complex structures).



Original source details: GELFAND-P000732

Uncertain reading. The parentheses say “all complex structures”. No further restriction is written here.

GELFAND-P000733

Atiyah.

(approach, did more)



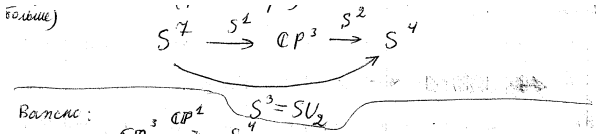
Original source details: GELFAND-P000733

Diagram and typed description.

$\mathbb{C}P^3 \xleftarrow{S^2} S^4$

The arrow in the upper line points left.

(Hopf fibr.)



Original source details: GELFAND-P000733

Diagram and typed description.

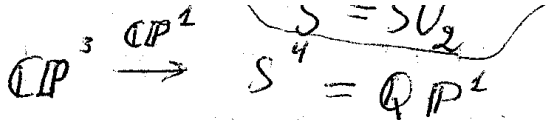
$$S^7 \xrightarrow{S^1} \mathbb{C}P^3 \xrightarrow{S^2} S^4$$
$$S^3 = SU_2$$

Important:



Original source details: GELFAND-P000733

Uncertain reading. The heading tentatively reads “Важно” (Important).



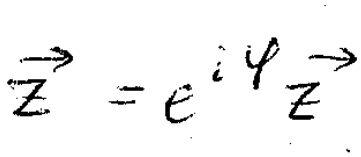
Original source details: GELFAND-P000733

Diagram and typed description.

$$\mathbb{C}P^3 \xrightarrow{\mathbb{C}P^1} S^4 = \mathbb{Q}P^1$$

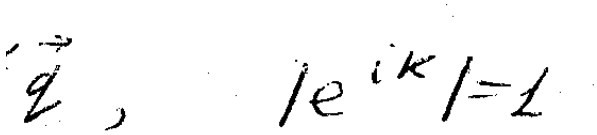
q_1, q_2 — quaternions.

$$q_1 = (z_1, z_2), \qquad q_2 = (z_3, z_4).$$
$$\vec{z} = e^{i\varphi}\vec{z}, \qquad \vec{q} \to e^{ik}\vec{q}, \qquad |e^{ik}| = 1.$$



Original source details: GELFAND-P000733

Uncertain reading. There is an equals sign between $\vec{z}$ and the exponential.



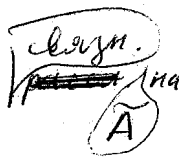
Original source details: GELFAND-P000733

Uncertain reading. After i in the exponent, the sign is tentatively k. It also resembles κ.

(unit quaternion; they form the grp. SU_2).

$$\left(\begin{array}{c|c} \underline{z_1, z_2} & \underline{z_3, z_4} \\ \text{quatern.} & \text{quatern.} \end{array}\right).$$

The conn. A on S^4 we lift to [crossed out: bund.?] [inserted above in a circle: conn.] [inserted below in a circle: $\widetilde{A}$] on $\mathbb{C}P^3$.

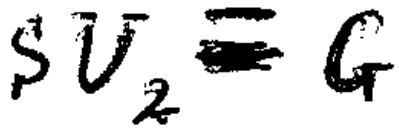


Original source details: GELFAND-P000733

Uncertain reading. The crossed-out word is tentatively “рассл.” (bund.). The insertions are circled.

Connection with grp. SU_2.

$$SU_2 = G$$



Uncertain reading. The sign before G tentatively reads $=$. The lower stroke is heavily retraced.

Original source details: GELFAND-P000733

The connection gives rise to an almost complex structure.

(the fibres are compl., and in the horiz. sps. it is lifted from the base).

Fibre $\mathbb{C}^2$.

Integrability of the almost compl. struct. $\Longleftrightarrow$ duality eqns..

$\Longleftrightarrow$ the ops. $\dfrac{\partial}{\partial z}$ commute. (repr. as a commuting pair)

Recapitulation.

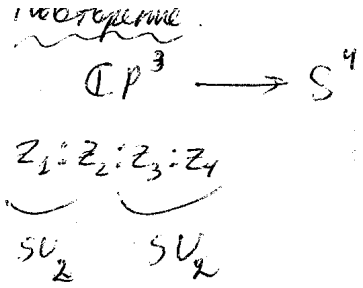


Diagram and typed description.

$$\mathbb{C}P^3 \longrightarrow S^4$$

$$\underbrace{z_1 : z_2}_{SU_2} : \underbrace{z_3 : z_4}_{SU_2}$$

Original source details: GELFAND-P000733

we act here and there by the same transfs. from SU_2.

[crossed out: $SU_2 \ldots \mathbb{C}P^3$?]



Uncertain reading. The crossed-out expression tentatively contains SU_2 and $\mathbb{C}P^3$. The sign between them is uncertain.

Original source details: GELFAND-P000733

$$\text{Fibre} = SU_2/(\text{diag. matrices}) = \mathbb{C}P^1.$$

There is a complex structure in the base and in the fibre.

Suppose on M^{2n} there is an almost compl. structure

(a way of making $\forall$ tangent sp. complex)

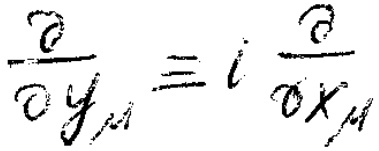
$$\partial_1, \ldots, \partial_n \text{ — a basis of vectors in the tangent sp.}$$
$$\partial_{n+1}, \ldots, \partial_{2n}$$

GELFAND-P000734

[source number: 71]

Let $\nabla_j = \partial_j + i\partial_{n+j}$

$$\frac{\partial}{\partial y_\mu} \equiv i \frac{\partial}{\partial x_\mu}$$



Uncertain reading. The manuscript uses a three-bar sign ≡ between the derivatives.

Original source details: GELFAND-P000734

Req. that they commute.

This is precisely integrability of the almost compl. structure.



Uncertain reading. The middle letters of the Russian abbreviation “квазикомпл.” (almost complex) are heavily retraced.

Original source details: GELFAND-P000734

GELFAND-P000735

15/XI-77. Gelfand Sem.

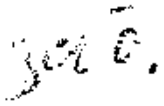
B. Sz.-Nagy.

A number of operator problems reduce to constructing contraction operators on Hardy spaces.



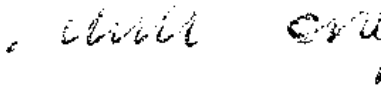
Uncertain reading. The faint first word tentatively reads “Ряд” (a number of).

Original source details: GELFAND-P000735



Uncertain reading. The second word tentatively reads “зад.” (problems).

Original source details: GELFAND-P000735



Uncertain reading. The abbreviation following “constructing” tentatively reads “сжат.” (contraction).

Original source details: GELFAND-P000735

H^∞ — bounded analytic functions in the disk.

[Crossed out] inner function

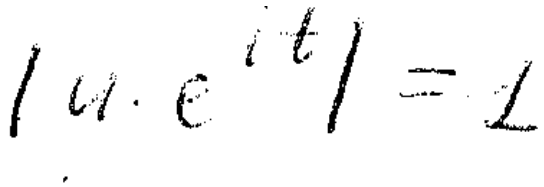
$$\underset{\text{def}}{=} \quad |u \cdot e^{it}| = 1$$

almost everywhere



Original source details: GELFAND-P000735

Uncertain reading. The crossed-out signs before “inner function” are illegible.



Original source details: GELFAND-P000735

Uncertain reading. The manuscript places a dot between u and e^{it}.

$$\bigwedge_{\alpha} u_\alpha$$

— greatest inner divisor of a family of functions



Original source details: GELFAND-P000735

Uncertain reading. A short curved stroke in the right margin. Its purpose is unclear.

Blaschke decompos.:

$$\forall \quad \text{[Crossed out]} \quad u = u_i u_o$$

(Smirnov thm.)



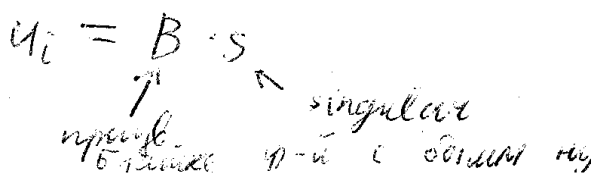
Original source details: GELFAND-P000735

Uncertain reading. The crossed-out signs between $\forall$ and u are illegible.



Original source details: GELFAND-P000735

Uncertain reading. The last subscript tentatively reads o.

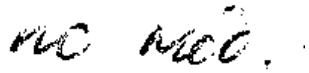


Original source details: GELFAND-P000735

Diagram and typed description.

$u_i = \underset{\text{Blaschke product}}{B} \cdot \underset{\text{singular}}{s}$

of functions with a common zero and by meas. [Crossed out].



Original source details: GELFAND-P000735

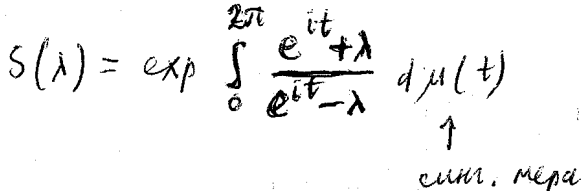
Uncertain reading. The last Russian abbreviation tentatively reads “мас.”. Its expansion is uncertain.



Original source details: GELFAND-P000735

Uncertain reading. The crossed-out group at the end of the line is illegible.

GELFAND-P000736



Original source details: GELFAND-P000736

Diagram and typed description.

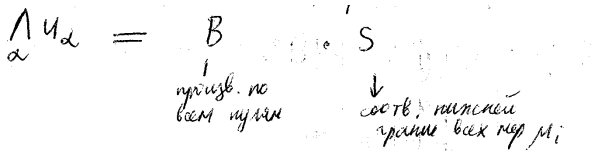
$$s(\lambda) = \exp \int_0^{2\pi} \frac{e^{it}+\lambda}{e^{it}-\lambda} \quad d\mu(t)$$

↑ sing. measure



Original source details: GELFAND-P000736

Uncertain reading. The initial sign tentatively reads s. Its case is uncertain.



Original source details: GELFAND-P000736

Diagram and typed description.

$$\bigwedge_\alpha u_\alpha = \quad B \quad \cdot \quad S$$

	↓	↓
	product over all zeros	corr. to the infimum of all measures μ_i



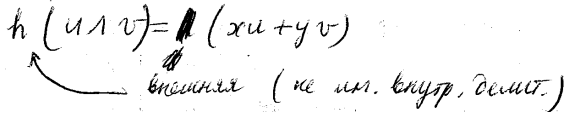
Original source details: GELFAND-P000736

Uncertain reading. The second factor tentatively reads capital S.

In the Hardy alg. it is false that

$$u \wedge v = xu + yv.$$

It is false even that



Original source details: GELFAND-P000736

Diagram and typed description.

$$h \quad (u \wedge v) = \quad \square \quad (xu + yv)$$

outer (having no inner divisor)



Original source details: GELFAND-P000736

Uncertain reading. The sign before the parentheses is blacked out. The stroke directly below it is also heavily crossed out.



Original source details: GELFAND-P000736

Uncertain reading. A broken curved stroke appears in the right margin. Its purpose is unclear.

Fix an inner fn. ω. Then $\forall u, v\ \exists x, y$:

$$xu + yv = h_\omega(u \wedge v), \quad \text{where } h_\omega \wedge \omega = 1$$

(but h_ω need not be outer)



Original source details: GELFAND-P000736

Uncertain reading. The last word tentatively reads “внешний” (outer). The middle letters are heavily retraced.

In fact, one can choose $x, y = \text{const}$

$$u + tv = h_t(u \wedge v)$$

M. Sherman (beautiful proofs of the lemma)

Generalization:

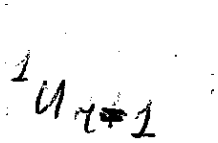
[crossed out] $\forall u_1, \dots, u_r,$



Original source details: GELFAND-P000736

Undeciphered passage. The overwritten group before the quantifier is illegible.

$$u_1 + tu_2 + \cdots + t^{r-1}u_{r+1} = h_t\Big(\bigwedge_k u_k\Big); \quad h_t \wedge \omega = 1$$



Original source details: GELFAND-P000736

Uncertain reading. The last subscript tentatively reads $r + 1$.



Original source details: GELFAND-P000736

Undeciphered passage. The earlier strokes beneath the heavy mark in the index are illegible.

(Except for a countable set of values of t)
and likewise for a finite number of rows
Then some elementary matrix transformations.
I. Krichever.
Rat. solutions of the Kadomtsev–Petviashvili equation:

$$\frac{3}{4}u_{yy} + \frac{\partial}{\partial x}\left\{[\text{crossed out}]\frac{\partial u}{\partial t} + \frac{1}{4}(6uu_x + u_{xxx})\right\} = 0 \tag{1}$$



Undeciphered passage. The crossed-out sign before the time derivative is illegible.

Original source details: GELFAND-P000736

GELFAND-P000737

[source number: 78]

Theorem. Seek solns. rational in x $u(x, y, t)$, decaying as $|x| \to \infty$. Then it is easy to see that



Uncertain reading. The word before “see” reads “легко” (easy). Its last letter is heavily retraced.

Original source details: GELFAND-P000737



Undeciphered passage. The earlier strokes of the last letter are illegible beneath the heavy retracing.

Original source details: GELFAND-P000737

$$u(x, y, t) = \sum \frac{-2}{\big(x - x_j(y, t)\big)^2}$$



Uncertain reading. The lower loop identifies the first argument of x_j as y.

Original source details: GELFAND-P000737



Undeciphered passage. The earlier strokes of the argument are illegible beneath the correction.

Original source details: GELFAND-P000737

1) The poles move in y acc. to Moser's syst. with Hamiltonian

$$H = \frac{1}{2}\sum p_j^2 + \sum_{i<j} \frac{1}{(x_i - x_j)^2}$$

(t fixed)

2) The poles move in t [crossed out] according to Hamiltonian F_3, one of the integrals of Moser's syst.



Undeciphered passage. The crossed-out mark after the abbreviation for "move" is illegible.

Original source details: GELFAND-P000737

We obtain 2 commuting flows determining [inserted above: (rat.)] solns. of eq. (1).



Diagram and typed description.

Original source details: GELFAND-P000737

$$(1) \Longleftrightarrow \left[\frac{\partial}{\partial y} - L_1, \frac{\partial}{\partial t} - L_2\right] = 0,$$

$$L_1 = \frac{d^2}{dx^2} + u, \qquad L_2 = \frac{d^3}{dx^3} + \frac{3}{2}u\frac{d}{dx} + u$$



Undeciphered passage. The coefficient reads $3/2$. The earlier strokes under the heavy retracing are illegible.

Original source details: GELFAND-P000737



Uncertain reading. The final term is written $+u$.

Original source details: GELFAND-P000737

We obtain commutative subalgebras in $\mathcal{D}_n$ (division ring of fractions of the ring of diff. ops. with polyn. coeffs. in n variables)

Next time Yu. I. Manin



Uncertain reading. The faint abbreviation before "раз" (time) tentatively reads "След." (next).

Original source details: GELFAND-P000737